BIOARCHAEOLOGICAL ANALYSIS OF FRAGMENTARY BURIALS FROM THE SITE OF JULCUY A LATE ARCHAIC SITE IN COASTAL ECUADOR

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Introduction

The archaeological site of Julcuy is a located along the Julcuy river, at the southern portion of the province of Manabí in coastal Ecuador. It was occupied relatively continuously from the earliest evidence of the Archaic period through to the later Manteño-Huancavilca culture. However, bioarchaeological analyses of the archaic period are rare as a result of preservation and excavation bias.

The Archaic period of Coastal Ecuador presents a blind spot for bioarchaeological analyses. Poor preservation conditions and the limited analysis of recovered human remains create gaps in our understanding and increase the risk of misinterpretation. The minimal public information on the individuals from the site of Julcuy have led to the pseudoscientific claims on the Julcuy Giant (Landon 2024). And yet these misinterpretations can be remedied through further study and a better understanding of the ancient peoples living in this region. This study, therefore, adds to the collected and shared data and we hope will spark further bioarchaeological investigations into this region and the peoples of the archaic period.

Individuals

- Excavations at the site in 2018 recovered two individuals from the lowest, stratum dating to the archaic period.
- Preservation of the remains at this riverbank site was poor as a result of the acidic soils, high rainfall, and moving water table.
- The remains of one of these individuals were cleaned prior to storage (Entierro 2) while the remains of the other were maintained in the matrix due to their fragility, fragmentation, and poor preservation (Entierro 1).
- Each burial was found in a flexed position facing up without associated cultural material.
- The individuals were buried in the same feature approximately 2 meters from each other
- Entierro 2 was recovered closer to the contemporary river, as they had been eroding out of the riverbank

Author Website

Poster

Results

Entierro 1:
- Size and density of skeletal elements suggest **small and gracile**.
- Most epiphyses were not found on the long bones, but this could be the result of erosion.
- Soil matrix and erosion obscure the area of the third molar, but all other permanent teeth had erupted and the anterior teeth that could be observed had complete or near complete apical closure
- Fusion of the epiphyses for olecranon process and the radial tubercle in addition to the closed apices suggest an age of **14-17 years.**
- Sex for an individual of that age from the cranium is unreliable and was therefore not attempted.
- Stature similarly is missing an adolescent reference population for stature from breadths and was therefore also not attempted.
- Within the burial were found faunal remains, primarily deer.

Entierro 2:
- Mostly complete, **robust**, and fairly preserved individual despite eroding from the riverbank.
- Missing mostly vertebral column and ribs, as well as facial bones.
- Complete fusion of long bone epiphyses, including the medial clavicle, suggest an individual over the age of **25 years**, but a lack of observable age characteristics of the pelvis, cranium, or teeth make reducing the age interval a challenge.
- **Male** sex was estimated using the Walker (2008) traits.
- Epiphyses had broken form the diaphysis and any metaphysis had eroded such that lengths could not be estimated.
- Stature was estimated to **161 ± 5.9 cm**.
- No faunal remains were found interred with this individual.

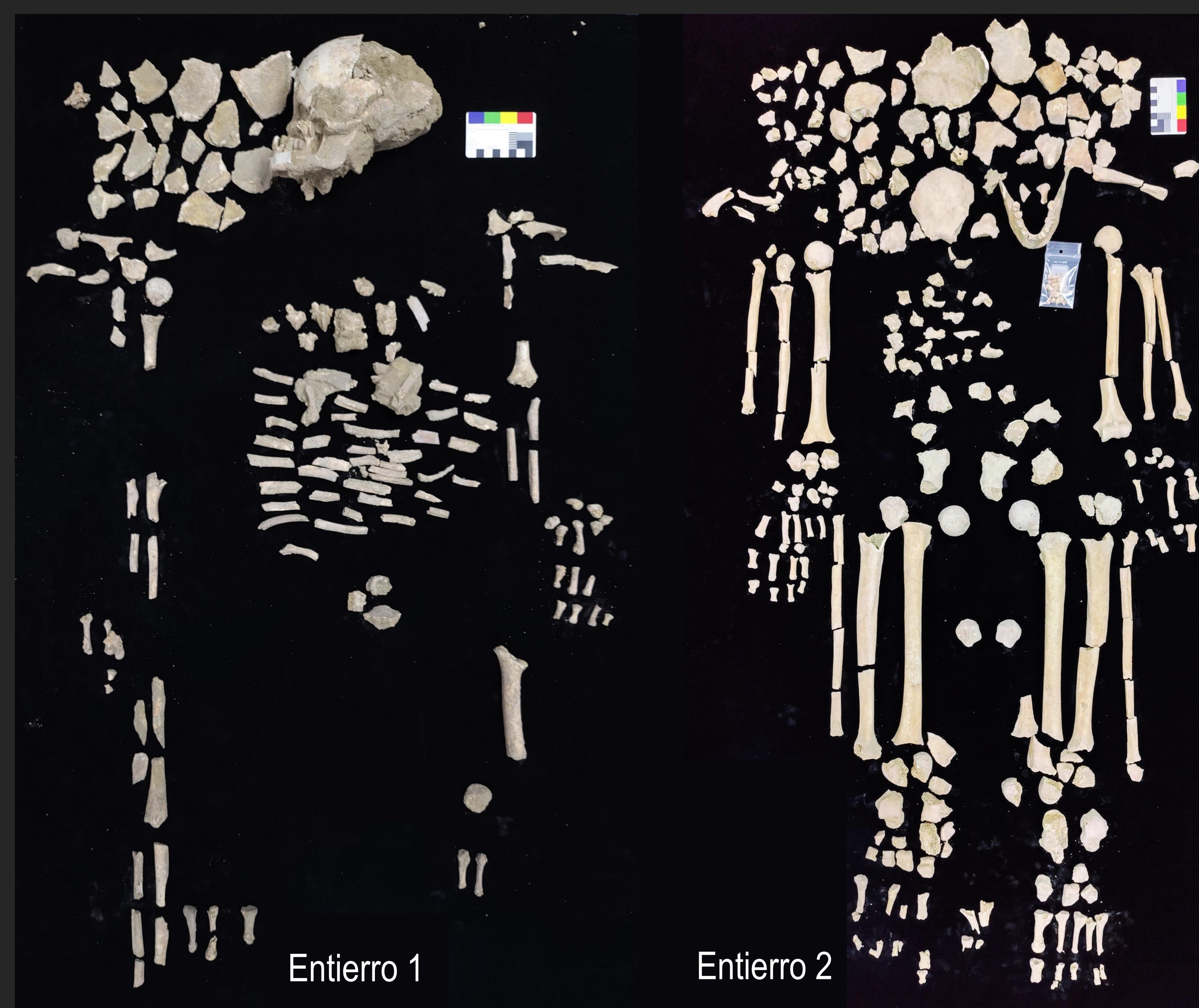

Fig. 1 – Entierro 1 and Entierro 2 recovered from the riverbank along the Julcuy River. Images of remains presented for educational purposes with the permission from the descendant communities and with respect to the individuals represented.

Fig. 2 – Location of the site of Julcuy in Coastal Ecuador.

Methods

- Morphological analyses of the individuals followed the standards published by Buikstra and Ubelaker (1994) and dental analyses from AlQhatani et al. (2010).
- Measurements were collected using the methods outlined in the Data Collections Protocol 2.0 (Langley et al. 2016)
- Age was estimated using fusion of epiphyses and apical root closures.
- Due to the fragmentary nature of the pelvis, Sex was estimated for Entierro 2 using cranial morphological features (Walker 2008). Entierrol 1 was found to be an adolescent and sex estimation was avoided due to reliability concerns.
- Stature was estimated for Entierro 2 using methods outlined in Anzellini and Toyne (2020), developed for ancient South American populations, and expanded to the use of breadths and widths of long bones from the same South American reference population.

Discussion

- Ecuador's archaic period is poorly understood due to a lack of investigation
- Some findings, like the Amantes de Sumpa, have added limited information.
- Pseudoarchaeological reports like those of the Julcuy Giant, which have been debunked through proper bioarchaeological analyses (Landol 2024), demonstrate the need for additional data about these ancient populations.
- The statures of Entierro 2 fits well within expected statures for native South Americans of the archaic period (Anzellini 2016).
- Entierro 1 and Entierro 2 were similarly interred, yet the younger individual had faunal remains from deer that may have played a funerary role.
- Due to the limited data resulting from poor preservation in the region, it is imperative for bioarchaeologists working in Coastal Ecuador to publish and share as much data and methods as possible.
- As we continue to collect bioarchaeological data, we may begin to piece together a much more cogent picture of the early settlement populations.

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